



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
REVIEW TEST
27-06-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TOAS & TNAS]

Time : 3 Hours

PAPER-1

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

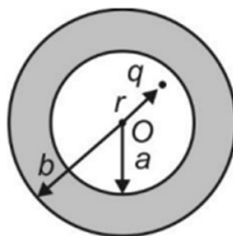
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. A point charge q is at a distance r from the centre O of an uncharged spherical conducting layer, whose inner and outer radii are equal to a and b respectively. If $r < a$, then the potential at the centre O is:



(A) $\frac{q}{4\pi\epsilon_0} \left(\frac{1}{r} - \frac{1}{a} + \frac{1}{b} \right)$

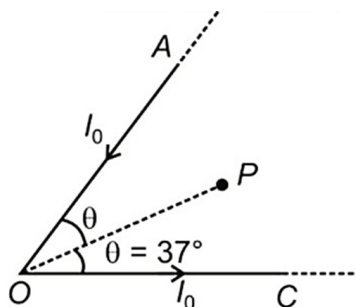
(B) $\frac{q}{4\pi\epsilon_0 r}$

(C) $\frac{q}{4\pi\epsilon_0} \left(\frac{1}{r} + \frac{1}{a} - \frac{1}{b} \right)$

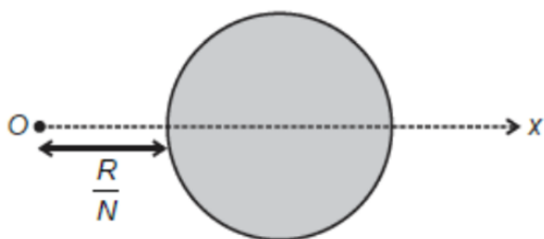
(D) $\frac{q}{4\pi\epsilon_0} \left(\frac{1}{a} + \frac{1}{b} - \frac{1}{r} \right)$

Space for Rough Work

2. Two wires A and C carry equal current $I_0 = 1\text{A}$ as shown in figure. One end of both wires extends to infinity. The magnitude of magnetic induction at point P on the angle bisector of AOC at a distance 1 cm from point O is:



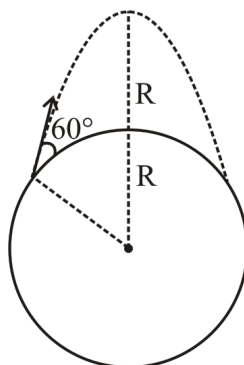
- (A) $3 \times 10^{-5}\text{T}$ (B) $6 \times 10^{-5}\text{T}$ (C) $4 \times 10^{-5}\text{T}$ (D) $8 \times 10^{-5}\text{T}$
3. A point object O is placed in front of a sphere of radius R and $\mu = \frac{3}{2}$ at distance $\frac{R}{N}$. Find value of N so that emergent rays are parallel to x-axis. Consider paraxial rays only:



- (A) 1 (B) 2 (C) 3 (D) 4

Space for Rough Work

4. A projectile is thrown at angle 60° with surface of earth, with velocity v_0 such that it attains a maximum height equal to radius of earth. Then :-



- (A) minimum velocity would be $\frac{V_0}{2}$ (B) minimum velocity would be $\frac{V_0}{4}$
 (C) minimum velocity would be $\frac{V_0}{3}$ (D) minimum velocity would be $\frac{V_0}{6}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. A thin non-conducting disc of radius R is uniformly charged with surface charge density σ . If potential at any point on the circumference of disc be V and total electrostatic potential energy of the disc be U . then:

(A) $V = \frac{\sigma R}{\pi \epsilon_0}$

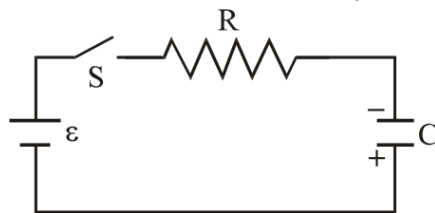
(B) $V = \frac{2\sigma R}{\pi \epsilon_0}$

(C) $U = \frac{2\sigma^2 R^3}{3\epsilon_0}$

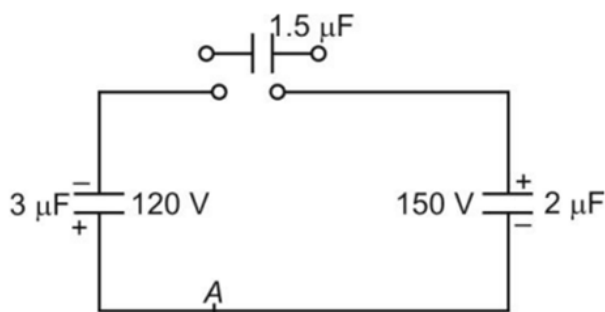
(D) $U = \frac{4}{3} \frac{\sigma^2 R^3}{\epsilon_0}$

Space for Rough Work

6. A capacitor C charged to a value $Q = \epsilon C$ is connected to a battery of emf ϵ as shown in the figure.



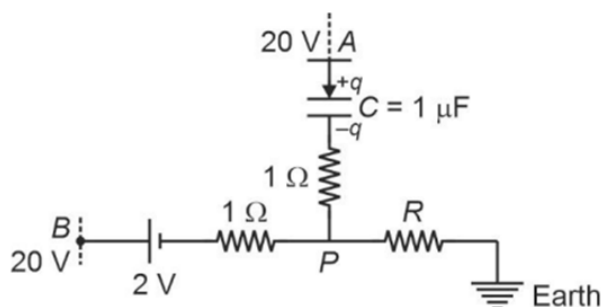
- (A) Charge on the capacitor is zero at time $RC \ln 2$ after the switch S is closed.
- (B) Charge on the capacitor is zero very long time after the switch S is closed.
- (C) Maximum charge on the capacitor during the process of charging is ϵC .
- (D) During the process of charging, the charging current first decreases and then increases exponentially.
7. Two capacitors of $2 \mu\text{F}$ and $3 \mu\text{F}$ are charged to 150 volt and 120 volt respectively. The plates of a capacitor are connected as shown in the figure. A discharged capacitor of capacity $1.5 \mu\text{F}$ falls to the free ends of the wire. Then after the system comes in steady state.



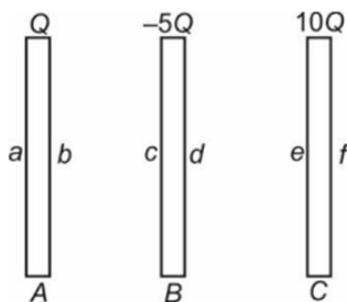
- (A) Charge on the $1.5 \mu\text{F}$ capacitor will become $180 \mu\text{C}$
- (B) Charge on the $2 \mu\text{F}$ capacitor will become $120 \mu\text{C}$
- (C) Positive charge flows through A from left to right
- (D) Positive charge flows through A from right to left

Space for Rough Work

8. The network shown in the figure is part of some bigger circuit. Charge on capacitor ($C = 1 \mu\text{F}$) at any time t is $q = 3(1 - e^{-t})$ in coulombs. If at $t = 0$ the potentials of point A and B is 20 V, then at $t = 0$ match the following.



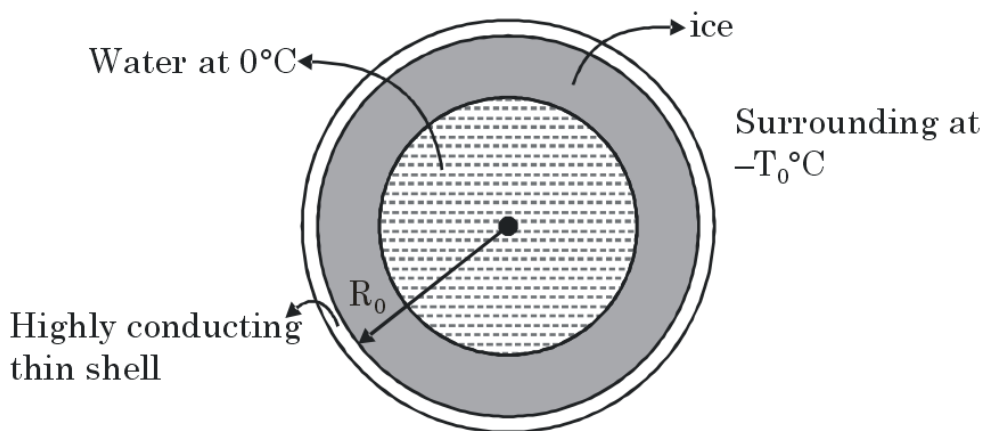
- (A) Current in branch AP is 3 A
(B) Current through R is 4 A
(C) Value of R is 4.25Ω
(D) Value of R is 1Ω
9. Three very large plates are given charges as shown in the figure. If the cross-sectional area of each plate is same, then the final charge distribution on the surface of the plates a, b, c, d, e, f is:



- (A) $7Q$ on surface e and $3Q$ on surface f
(B) $-2Q$ on surface b and $3Q$ on surface a
(C) $-7Q$ on surface d and $2Q$ on surface c
(D) The magnitude of charges at surface b, c, d, e is equal

Space for Rough Work

10. A highly thin conducting spherical shell of radius R_0 is completely filled with water at 0°C . Outside surrounding temperature is $-T_0^\circ\text{C}$. Latent heat of fusion of water is L_0 and thermal conductivity of ice is K_{ice} . Assume heat is flowing symmetrically in radial direction. Choose the CORRECT option(s). Density of water is ρ_0 . Neglect the expansion of water on freezing.



- (A) Time in which complete water will freeze, is given by $\frac{\rho_0 L_0 R_0^2}{6K_{\text{ice}} T_0}$
- (B) Time in which complete water will freeze, is given by $\frac{\rho_0 L_0 R_0^2}{4K_{\text{ice}} T_0}$
- (C) Time in which thickness of ice will become $\frac{R_0}{2}$, is given by $\frac{\rho_0 L_0 R_0^2}{12K_{\text{ice}} T_0}$
- (D) Total heat released by water in freezing, is given by $\frac{4\pi\rho_0 L_0 R_0^3}{3}$

Space for Rough Work

11. A parallel plate capacitor has a dielectric slab of dielectric constant K between its plates that cover $1/3$ of the area of its plates, as shown in the figure. The total capacitance of the capacitor is C while that of the portion with dielectric in between is C_2 . When the capacitor is charged the plane area covered by the dielectric gets charge Q_1 and the rest of the area gets charge Q_2 . The electric field in the dielectric is E_1 and that in the other portion is E_2 . Choose the correct option(s), ignoring edge effects.



- (A) $\frac{E_1}{E_2} = K$ (B) $\frac{E_1}{E_2} = \frac{1}{K}$ (C) $\frac{Q_1}{Q_2} = \frac{3}{K}$ (D) $\frac{C}{C_2} = \frac{2+K}{K}$
12. A particular spherical capacitor consists of two concentric conducting spherical shells of radii R and $2R$. Neglect edge effects in each.
- (A) A coaxial cylindrical capacitor, made of two concentric cylindrical conductors of radii R and $2R$, has the same capacitance as the spherical capacitor. Its length should be $4R/\ln 2$
- (B) A coaxial cylindrical capacitor, made of two concentric cylindrical conductors of radii R and $2R$, has the same capacitance as the spherical capacitor. Its length should be $4R \ln 2$
- (C) A parallel-plate capacitor, made of two circular plates of radius R , has the same capacitance as the spherical capacitor. The plates are at a distance of $R/8$.
- (D) A parallel-plate capacitor, made of two circular plates of radius R , has the same capacitance as the spherical capacitor. The plates are at a distance of $R/4$.

Space for Rough Work

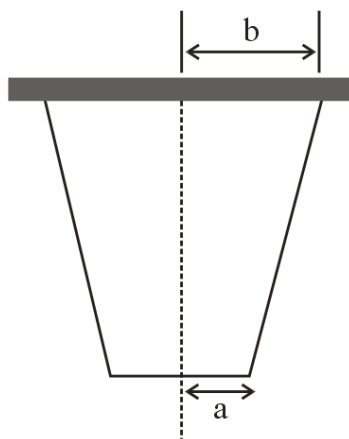
SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

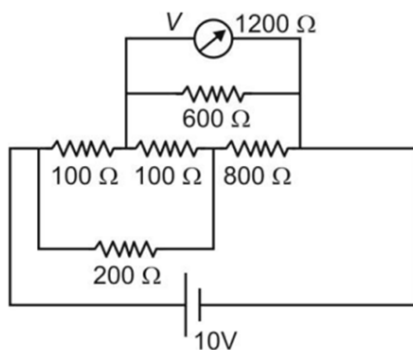
Zero Marks : 0 In all other cases.

1. The current I flows through an equilateral triangular loop of side length $\frac{x}{8}$. If magnetic field at the centroid of the triangular loop is $\frac{N\mu_0 I}{\pi x}$, then find the value of $N/6$.
2. A body of mass 3.14 kg is suspended from one end of a wire of length 10m. The radius of the wire is changing uniformly from 10×10^{-4} m at one end to 5×10^{-4} m at the other end. Find the change in length of the wire in mm. Young's modulus of the material of the wire is 2×10^{11} N/m². (mass of wire is negligible) Take : $\pi = 3.14$ and $g = 10$ m/s²



Space for Rough Work

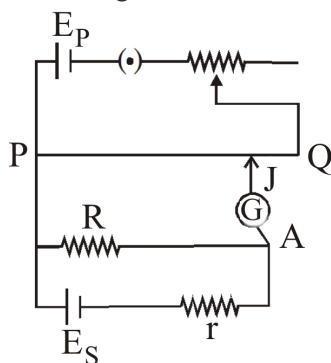
3. Calculate the reading of voltmeter (in volts) in the circuit shown below



4. A highly conducting uniform sphere of thermal capacity C is heated by an electric heater, a resistance R fitted within the sphere. A constant current I is passed through the heater starting at time $t = 0$ which gives constant power. The sphere loses heat at a rate equal to k times the temperature difference between the sphere and the surrounding. The initial temperature of the sphere and that of the surrounding is 0°C . the time at which sphere attains half of its maximum attainable temperature (in $^\circ\text{C}$) is $\frac{C}{k} \ln a$. Then find the value of "a".

Space for Rough Work

5. A body of mass 2 gm has specific heat capacity $0.25 \text{ cal/gm-}^\circ\text{C}$ and surface area 30 cm^2 . It is kept in an enclosure which is maintained at temperature 127°C . Body is being supplied heat at rate 2 watt. If temperature of body is 27°C and emissivity of body is 0.5. Determine the rate at which its temperature is changing (in K/s) at the given instant. (Stefan's constant is $5.67 \times 10^{-8} \text{ W/m}^2\text{-K}^4$. $1 \text{ cal} = 4.2 \text{ J}$)
6. One of the circuits for the measurement of resistance by potentiometer is shown. The galvanometer is connected at point A and zero deflection is observed at length $PJ = 30 \text{ cm}$. In second case the secondary cell is changed. Take $E_S = 10 \text{ V}$ and $r = 1\Omega$ in 1st reading and $E_S = 5\text{V}$ and $r = 2\Omega$ in 2nd reading. In second case, the zero deflection is observed at length $PJ = 10 \text{ cm}$. What is the resistance R (in ohm) ?



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

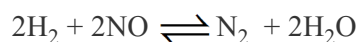
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

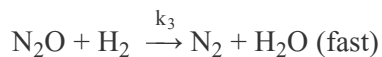
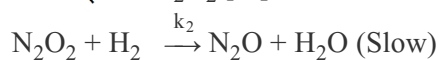
-
1. For a first order reaction $A \rightarrow B + C$ all the reactant and product molecules are in gaseous state. If the initial pressure of the reactant molecules is 100mm and after 30 minutes the total pressure of the mixture containing molecules of all three A, B & C gases is 140 mm, then find the time interval at which the initial pressure of A will be reduced to 50 mm. ($\log_{10} 3 = 0.5$, $\log_{10} 2 = 0.3$)
- (A) 45 min (B) 30 min (C) 60 min (D) 15 min
-

Space for Rough Work

2. For the reaction:



the following mechanism has been suggested :



According to above mechanism which option is correctly matched:

(A) $-\frac{1}{2} \frac{d[\text{H}_2]}{dt} = K[\text{NO}] [\text{H}_2]$ Where $K = k_2 \times k_3$

(B) $-\frac{1}{2} \frac{d[\text{NO}]}{dt} = K[\text{NO}] [\text{H}_2]$ Where $K = k_2 \times K_1$

(C) $\frac{d[\text{N}_2]}{dt} = K[\text{NO}] [\text{H}_2]$ Where $K = k_2 \times K_1$

(D) $-\frac{1}{2} \frac{d[\text{NO}]}{dt} = K[\text{NO}]^2 [\text{H}_2]$ Where $K = k_2 \times K_1$

Space for Rough Work

3. In the cyanide extraction process of silver from argentite ore, the oxidizing and reducing agents used are
- (A) O_2 and CO respectively
(B) O_2 and Zn dust respectively
(C) HNO_3 and Zn dust respectively
(D) HNO_3 and CO respectively
4. Compare strength of H-bond among the following :
- (P) $H - F \cdots \cdots H - F$
(Q) $H - O \cdots \cdots H - O$
(R) $H - N \cdots \cdots H - N$
- (A) $P > Q > R$ (B) $P < Q < R$ (C) $P > R > Q$ (D) $R > P > Q$
-

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

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- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. Which of the following is/are true regarding FCC unit cell ?

(a = edge length of unit cell, r = radius of lattice point)

(A) The smallest distance between an octahedral void and tetrahedral void is $\frac{\sqrt{3}}{4}a$.

(B) The smallest distance between two octahedral voids is $\frac{a}{\sqrt{2}}$.

(C) The smallest distance between two tetrahedral voids is $\frac{\sqrt{3}}{2}a$.

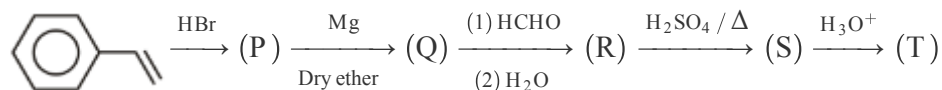
(D) The smallest distance between planes of two successive layers in ABC ABC ___type packing is $\sqrt{\frac{8}{3}}r$.

Space for Rough Work

6. The wave function for an electron in H-like atom is given as $\psi(r, \theta, \phi) = \frac{1}{4\sqrt{2\pi}} \cdot \left(\frac{Z}{a_0}\right)^{5/2} \cdot r \cdot e^{-Zr/2a_0} \cdot \cos \theta$. The correct information(s) about this electron is/are -

- (A) It belongs to a 2p orbital
- (B) This orbital has no radial node
- (C) It belongs to a 3p orbital
- (D) Total nodes for the given orbital is 1.

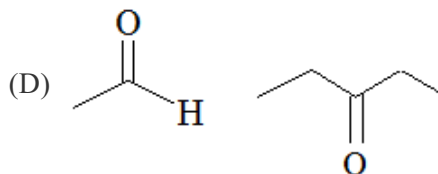
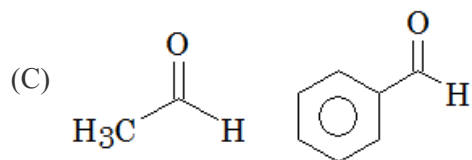
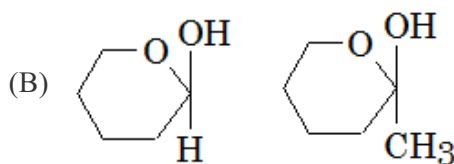
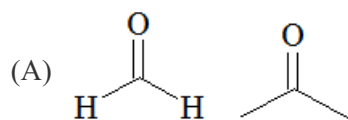
7. For the following reaction the correct option is/are :



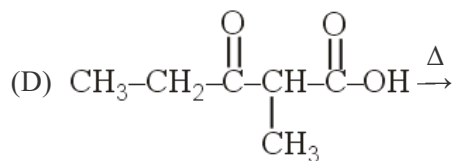
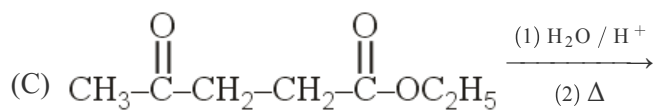
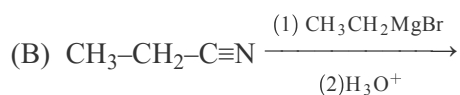
- (A) Both (R) & (T) gives immediate turbidity with HCl + anhy ZnCl_2
- (B) (P) shows optical isomerism
- (C) (S) shows geometrical isomerism
- (D) (Q) is known as grignard reagent

Space for Rough Work

8. Among the following pair of compounds, identify the pair(s) which can be differentiated by Fehling as well as Tollen's reagent ?



9. Which of the following will give 3-pentanone.



Space for Rough Work

10. Select the **CORRECT** matches

Column-I (Change)	Column-II (Given change involve)
(A) $\text{ZnO} \rightarrow \text{Zn}$	$\Rightarrow$ Self reduction
(B) $\text{Cu}(\text{impure}) \rightarrow \text{Cu}(\text{pure})$	$\Rightarrow$ Electrefining
(C) Pig iron $\rightarrow$ wrought iron	$\Rightarrow$ Impurity removed by oxidation method
(D) $\text{PbS} \rightarrow \text{PbO}$	$\Rightarrow$ Calcination
(A) A	(B) B
(C) C	(D) D

11. Which one of the following is correctly matched ?

(A) $2\text{Fe} + \text{Al}_2\text{O}_3 \rightarrow 2\text{Al} + \text{Fe}_2\text{O}_3$	Thermite process for extraction of iron
(B) $\text{Ti} + 2\text{I}_2 \xrightarrow{523\text{K}} \text{TiI}_4 \xrightarrow{1700\text{K}} \text{Ti} + 2\text{I}_2$	Mond process
(C) $\text{Ni} + 4\text{CO} \xrightarrow{\text{low temperature}} \text{Ni}(\text{CO})_4 \xrightarrow{\text{high temperature}} \text{Ni} + 4\text{CO}$	Van Arkel process
(D) $\text{Cu}_2\text{S} + 2\text{Cu}_2\text{O} \rightarrow 6\text{Cu} + \text{SO}_2$	Self reduction method
(A) (A)	(B) (B)
(C) (C)	(D) (D)

12. Which of the following anionic radicals are identified by diluted H_2SO_4 :-

(A) NO_3^-	(B) SO_3^{2-}
(C) Cl^-	(D) CO_3^{2-}

Space for Rough Work

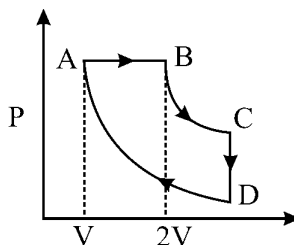
SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

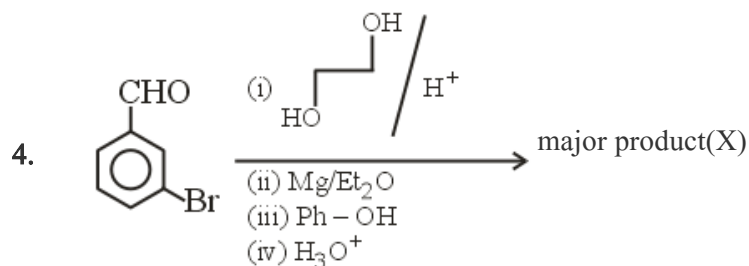
Zero Marks : 0 In all other cases.

1. In a polytropic expansion process ($PV^x = \text{constant}$), a diatomic ideal gas does an expansion work of $\frac{y}{2}$ Joules by absorbing 'y' Joules heat from surrounding. The value of 'x' is-
2. Calculate magnitude of net work in following cyclic process for 1 mole of ideal monoatomic gas (in cal.)
(Given - $B \rightarrow C$ - isothermal process, $D \rightarrow A$ - adiabatic process, $T_A = 300$ K
 $T_D = 75$ K, $R = 2$ cal/mole-K, $\ln 2 = 0.7$)



3. Consider all possible isomeric ketones including stereoisomers of molecular formula $C_6H_{12}O$, All these isomers are independently reacted with $NaBH_4$ (**NOTE** : stereoisomers are also reacted separately). The total number of ketones that give a racemic product(s) is/are

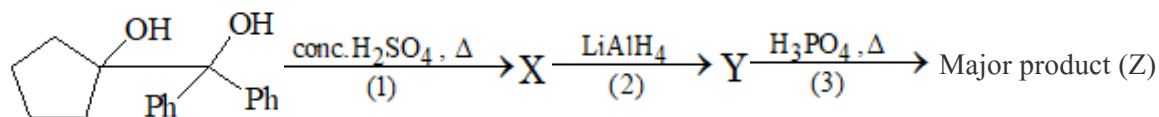
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How many statement is/are correct for [X]

- (a) It can give orange colour compound with 2,4-D.N.P.
- (b) It can give positive Silver mirror test.
- (c) It can be oxidised by Feheling solution
- (d) It can give yellow ppt. with NaOI, Δ
- (e) It can be identified by NaHSO₃

5. Observe the following reaction sequence



Calculate molecular mass (W) of product Z and report your answer as N, where $N = \frac{W}{3}$

6. In Industrial extraction, the number of metals produced by reduction of their oxides with carbon or with more active metal among the following is :

Fe, Li, Mg, Al, Au, Na, Sn, Zn

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Let $f(x) = \begin{cases} x^2 e^{2(x-1)}; & 0 \leq x \leq 1 \\ a \operatorname{sgn}(x+1) \cos(2x-2) + bx^2, & 1 < x \leq 2 \end{cases}$. If $f(x)$ is differentiable at $x = 1$, then
(where $\operatorname{sgn}(x)$ denotes signum function of x)

(A) $a = -1, b = 2$ (B) $a = -3, b = -2$ (C) $a = -3, b = 4$ (D) $a = -5, b = -4$

2. The value of the definite integral $\int_{-\infty}^a \frac{(\sin^{-1} e^x + \sec^{-1} e^{-x}) dx}{(\tan^{-1} e^a + \tan^{-1} e^x)(e^x + e^{-x})}$ ($a \in \mathbb{R}$) is :

(A) $\ln 3$ (B) dependent on a
(C) $\frac{\pi}{2} \ln 2$ (D) $\frac{\pi}{2} \ln(2 \tan^{-1} e^a)$

Space for Rough Work

3. If $y = \sin^{-1}(2x\sqrt{1-x^2}) + \sec^{-1} \frac{1}{\sqrt{1-x^2}}$, then $\frac{dy}{dx}$ can be
- (A) $\frac{5}{\sqrt{1-x^2}}$ (B) $\frac{2}{\sqrt{1-x^2}}$
- (C) $\frac{3}{\sqrt{1-x^2}}$ (D) $\frac{1}{\sqrt{1+x^2}}$
4. From point P(0, 0) pair of tangents PA and PB are drawn to the circle $x^2 + y^2 + 10x + 10y + 25 = 0$ and a variable tangent at a point L in minor arc AB (other than A and B) meet PA and PB at M and N respectively. The angle which MN subtends at the centre of the circle, is
- (A) constant for all positions of L
- (B) less than 45° for all position of L
- (C) more than 45° for all position of L
- (D) can be less than 45° for some positions of L

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. If $\int \sqrt{\sec^2 x + 3} dx = f(x) + \sqrt{3} g(x) + c$ where c is constant of integration and $f(0) = \ln 2$, $g\left(\frac{\pi}{2}\right) = \frac{\pi}{3} + 1$, then

(A) $f(x)$ is an odd function

(B) $f\left(\frac{\pi}{4}\right) = \ln(1 + \sqrt{5})$

(C) $g\left(\frac{3\pi}{2}\right) + g\left(\frac{\pi}{2}\right) = 2$

(D) $g(x)$ is an odd function

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6. If $I = \int_{-\pi/4}^{\pi/4} \frac{e^{\sec x} |\sin x| \sec^2 x}{(1 + e^{\sin x})} dx$, then

(A) I is an irrational number

(B) $I = e^{\sqrt{2}} - e$

(C) $I = 3.72$

(D) I is a rational number

7. If $f(x)$ is differentiable at $x = 5$ and $f(5) = 1$, $f'(5) = \frac{1}{12e}$, then value of $\lim_{x \rightarrow 5} (x - 5) \int_1^{f^2(x)} \frac{(x + t) e^t dt}{(x - 5)^2}$ is

(A) 0

(B) 1

(C) $\frac{5}{4}$

(D) $\frac{5}{6}$

8. Let a point 'R' be between P(1,0) and Q(e,1) on the graph of $y = \ln x$ such that the tangent to the graph at point 'R' is parallel to the line through points P and Q then which option is/are correct -

(A) x co-ordinate of the point 'R' is $e - 1$

(B) x co-ordinate of the point 'R' is $\frac{1}{e - 1}$

(C) if the equation of tangent at R is $y = f(x)$ then $\int_1^e f(x) dx > \int_1^e \ln x dx$

(D) normal at P for $y = \ln x$ intersect $y = e^x$ at (0, 1)

Space for Rough Work

9. If $f(x)$ be an identity function in $\mathbb{R}$ and $g(x) = \sum_{k=1}^3 \{f(x) - (2016 + k)\}^{-1}$, then

- (A) $g(x)$ is strictly decreasing in $(2018, 2019)$
 (B) $g(x)$ has two distinct real roots
 (C) Slope of tangent to the curve $g(x)$ at $x = f(2016)$ is $-\frac{49}{36}$
 (D) $\lim_{x \rightarrow -\infty} g(x) = 0$

10. Let $f, g : [-2, 3] \rightarrow \mathbb{R}$ be continuous function which are twice differentiable on the interval $(-2, 3)$. Let the values of f and g at the points $-2, 0, 1$ and 3 be as given in the following table :

	$x = -2$	$x = 0$	$x = 1$	$x = 3$
$f(x)$	11	-4	6	16
$g(x)$	1	-2	0	2

In each of the intervals $(-2, 0)$, $(0, 1)$ and $(1, 3)$ the function $(f - 5g)''$ never vanishes. Then the correct statement(s) is (are)

- (A) $f'(x) - 5g'(x) = 0$ has exactly three solutions in $(-2, 0) \cup (0, 1) \cup (1, 3)$
 (B) $f'(x) - 5g'(x) = 0$ has exactly one solutions in $(0, 1)$
 (C) $f'(x) - 5g'(x) = 0$ has at least one solution in $(1, 3)$
 (D) $f'(x) - 5g'(x) = 0$ has at least two solution in $(-2, 0)$

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11. Let $f(\theta) = \frac{1}{\tan^9 \theta} \left((1 + \tan \theta)^{10} + (2 + \tan \theta)^{10} + \dots + (20 + \tan \theta)^{10} \right) - 20 \tan \theta$. If $\lim_{\theta \rightarrow \frac{\pi}{2}^-} f(\theta)$ is ℓ , then
- (A) ℓ is divisible by 100 (B) ℓ is divisible by 7
 (C) ℓ is divisible by 30 (D) ℓ is a prime number
12. $f_1(x) = \sin^{-1}(\cos(\sin^2 x))$, $f_3(x) = \sin^{-1}(\cos(\cos^2 x))$
 $f_2(x) = \cos^{-1}(\sin(\cos^2 x))$, $f_4(x) = \cos^{-1}(\sin(\sin^2 x))$ then which of the following is correct ?
- (A) $\sum_{i=1}^4 f_i(x) = 2\pi - 2$ (B) $\sum_{i=1}^4 (-1)^i f_i(x) = 0$
 (C) $\sum_{i=1}^4 (-1)^{\left[\frac{i}{3}\right]} f_i(x) = 0$ (D) $f_1(x) > f_2(x) \Rightarrow \cos 2x < 0$

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SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
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- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. If $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \frac{3x^2 + mx + n}{x^2 + 1}$ has range $[-4, 3)$ then $m^2 + n^2$ is
2. The number of integral values of k for which the equation $e^x = \frac{k}{x-3}$, has exactly two solutions is-
3. If $\int \frac{(3x^2 + 2x)dx}{x^6 + 2x^5 + x^4 + 2x^3 + 2x^2 + 5} = \frac{1}{2} \tan^{-1}(g(x)) + C$, then the value of $(g(1) + g(0))$ is

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4. Let $f(x)$ be a twice differentiable function satisfying $xf(x^2) + (x^2 + 1)f(x) = x$, $\forall x \in \mathbb{R}$ then the value of $|f''(1)|$ is
5. If $f(x+1) = e^{x^2}f(x)$, where $f(x)$ is differentiable function such that $f'\left(\frac{1}{2}\right) = 5f\left(\frac{1}{2}\right)$, then the value of $\frac{f'\left(N + \frac{1}{2}\right)}{f\left(N + \frac{1}{2}\right)} - N^2$ is ($N = 1, 2, \dots$)
6. If the solution set of $\sin^{-1}x + \sin^{-1}y \geq (x+y)^2 - 2[x(y+1) - (1-y)] + \pi$ is (x_0, y_0) , then $3x_0 + y_0$ is

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Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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